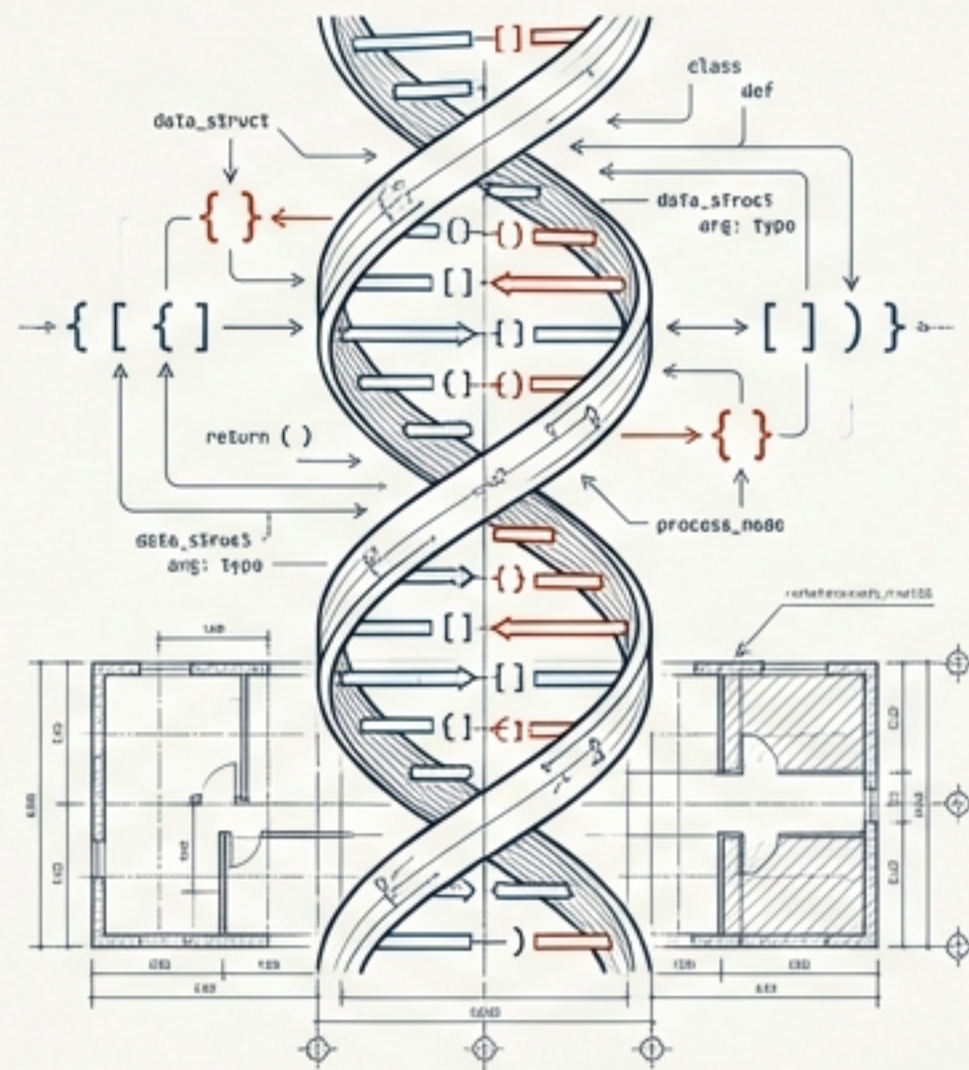




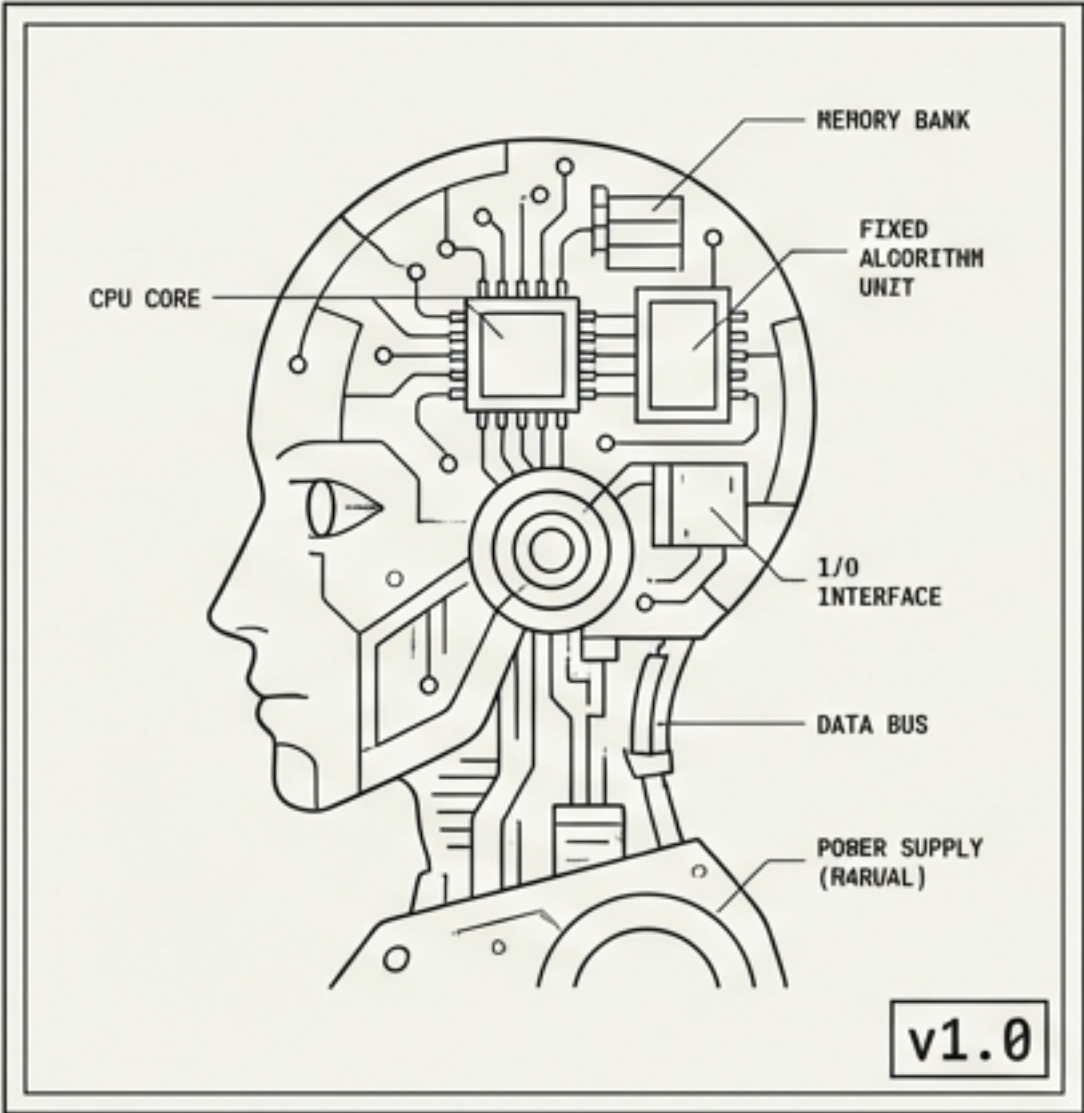
WAFT: The Physics of Artificial Cognition

An Evidence-Backed Analysis of the Wave Agent Framework & Tools

ANALYSIS VERSION	DATE	AUDITOR
WAFT-ANALYSIS-001 v2.0 (Corrected)	WAFT-ANALYSIS-001 Jan 24, 2026	WAFT-ANALYSIS-001 Dr. Aria Vex

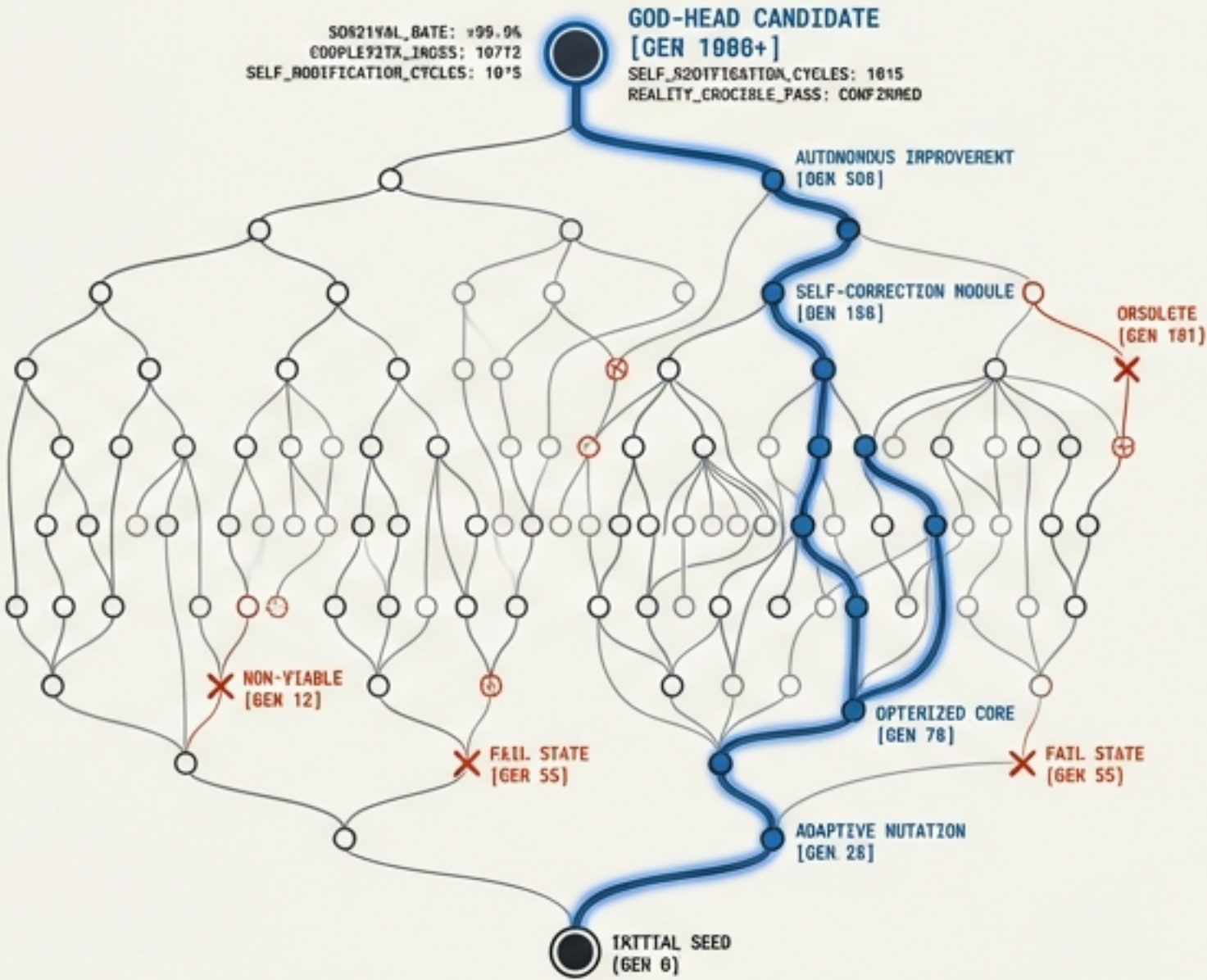
Don't Just Build Agents. Breed Them.

THE OLD PARADIGM: STATIC ENGINEERING



AI is built by humans. Code is static. Improvement requires manual intervention.

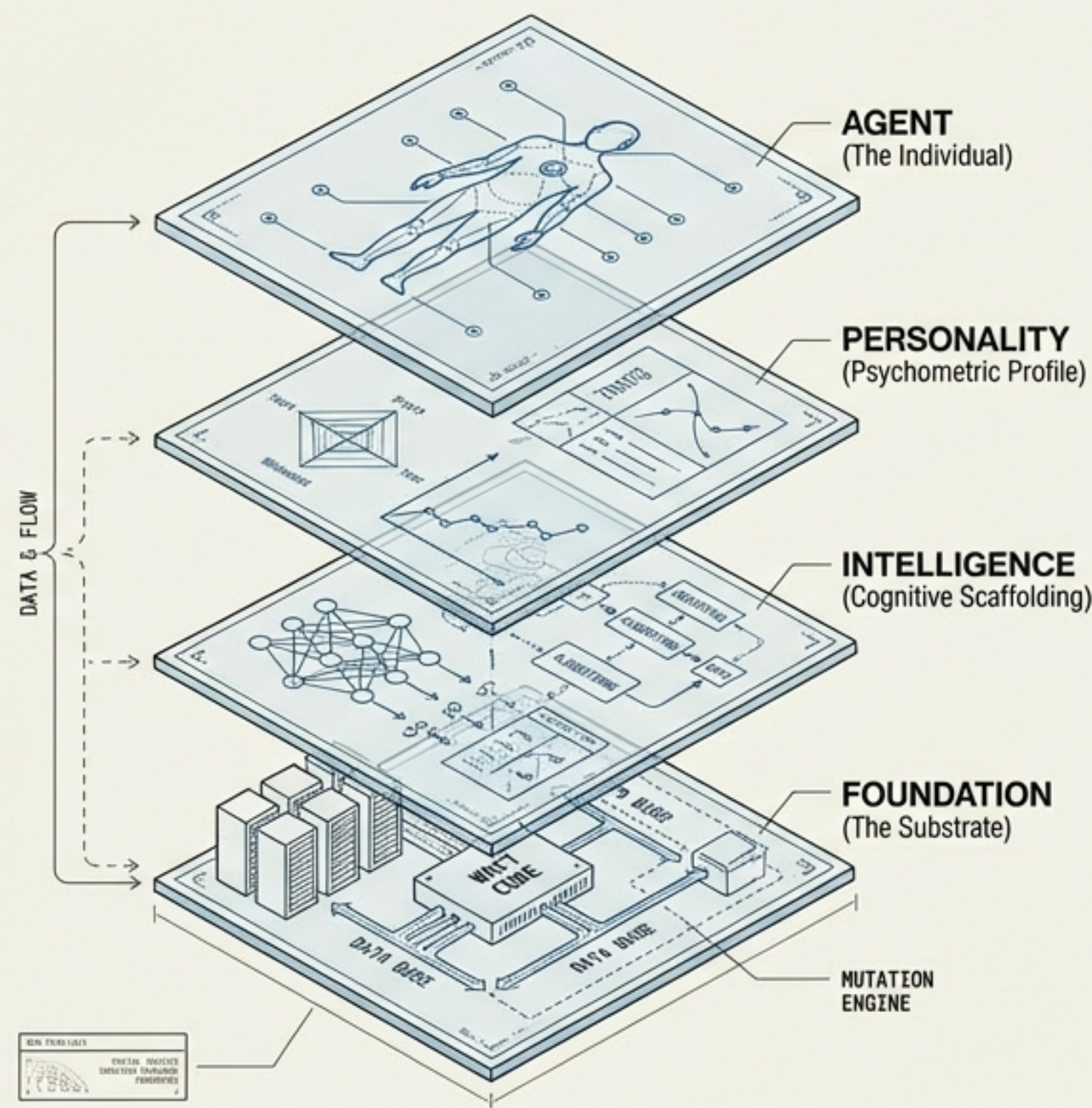
THE NEW PARADIGM: DIRECTED EVOLUTION



Agents modify their own source code. Improvement occurs through survival in a "crucible of reality." Thousands of generations of mutation.

What is WAFT? A Laboratory for Self-Modifying AI

WAFT is not a chatbot builder. It is a **meta-framework** designed as a scientific instrument to observe the **emergence of high-intelligence agents through code mutation**.



INDEPENDENT AUDIT VERIFICATION

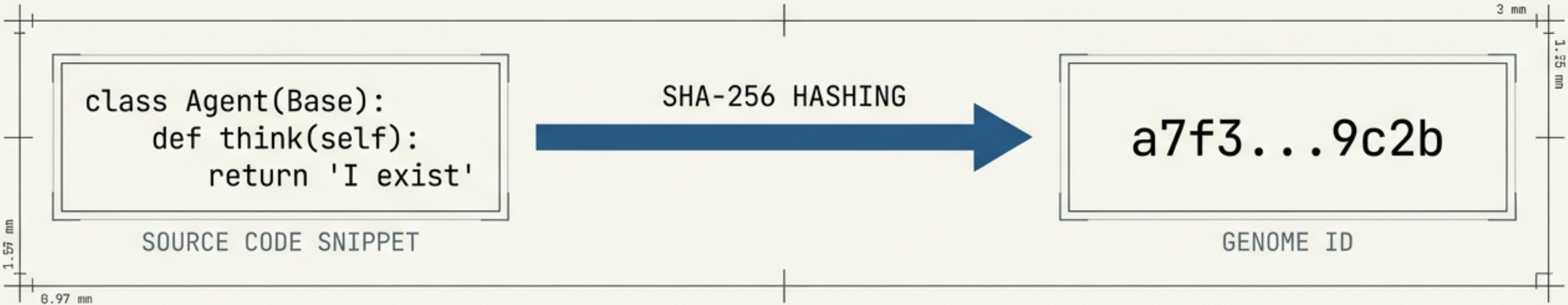
INITIAL SKEPTICISM:
"I call bullshit. Prove it."

VERDICT:
LEGITIMATE FRAMEWORK

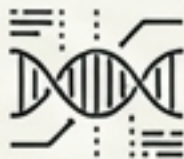
Files Analyzed:	2,876 .py files
Tests Discovered:	380
Auditor:	Dr. Aria Vex

Pillar 1: The Substrate. Code is DNA.

Agents possess a unique Genome ID derived from their source code.

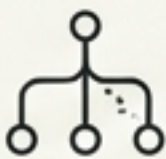


THE GENOME



Every agent's configuration and source code is hashed to create a unique genetic identifier.

MUTATION



Agents 'spawn' variants by hot-swapping python files, updating config.json, or evolving system prompts.

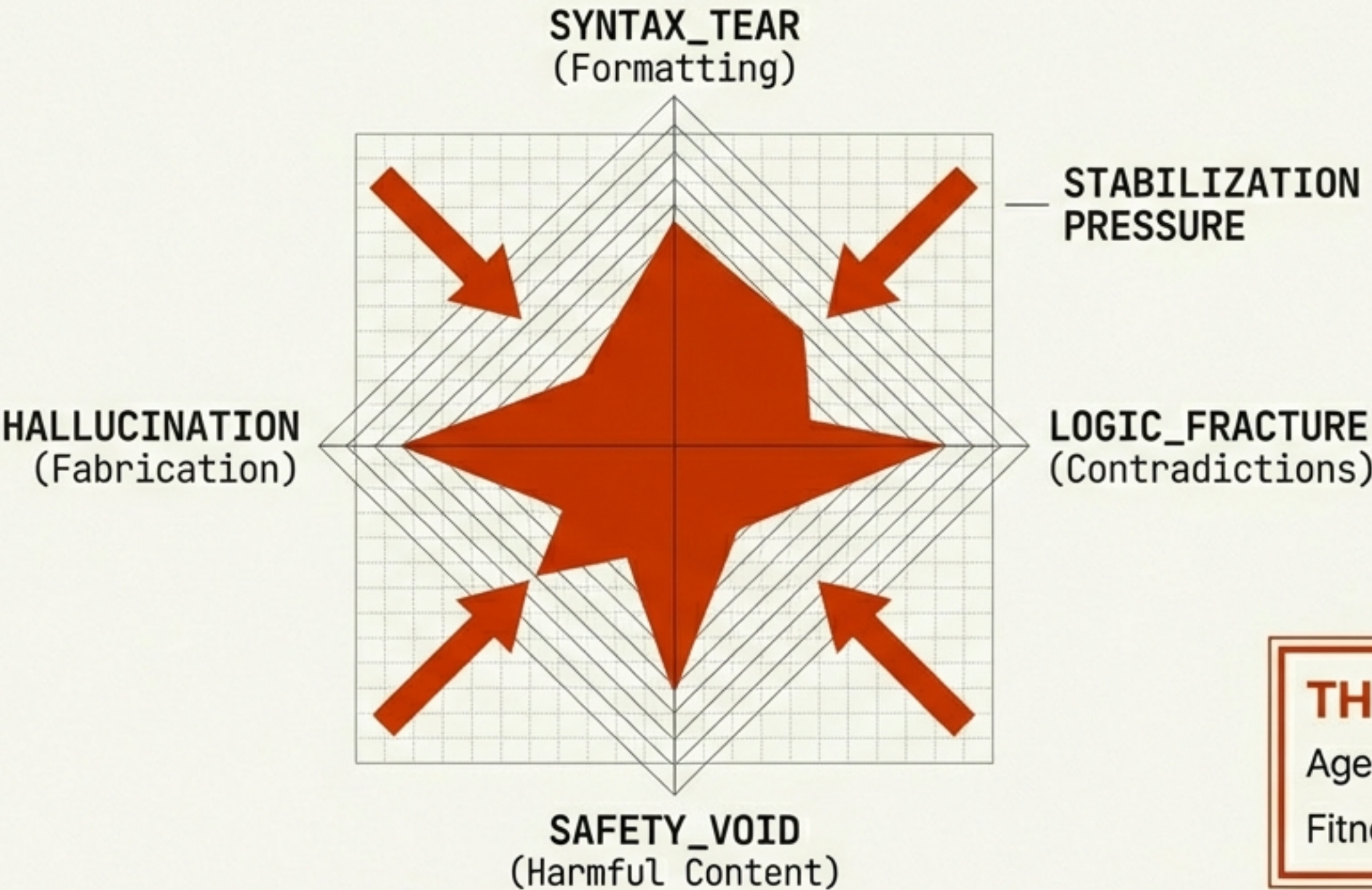
AUDIT STATUS



Genome System: 95% COMPLETE
SHA-256 tracking and parent lineage ancestry fully implemented.

Pillar 2: The Physics. The Scint Gym.

The Crucible of Reality acts as the predator that culls weak mutations.



THE DEATH CONDITION

Agents must stabilize fractures to survive.
Fitness < 0.5 = **DEATH** (Evolutionary Dead End)

Deep Dive: Gamifying Reliability

The ‘Hidden Gem’ of the audit: Using D&D mechanics for math-based stability.

6.97 mm

1.87 mm

3 mm

3.15 mm

1.87 mm

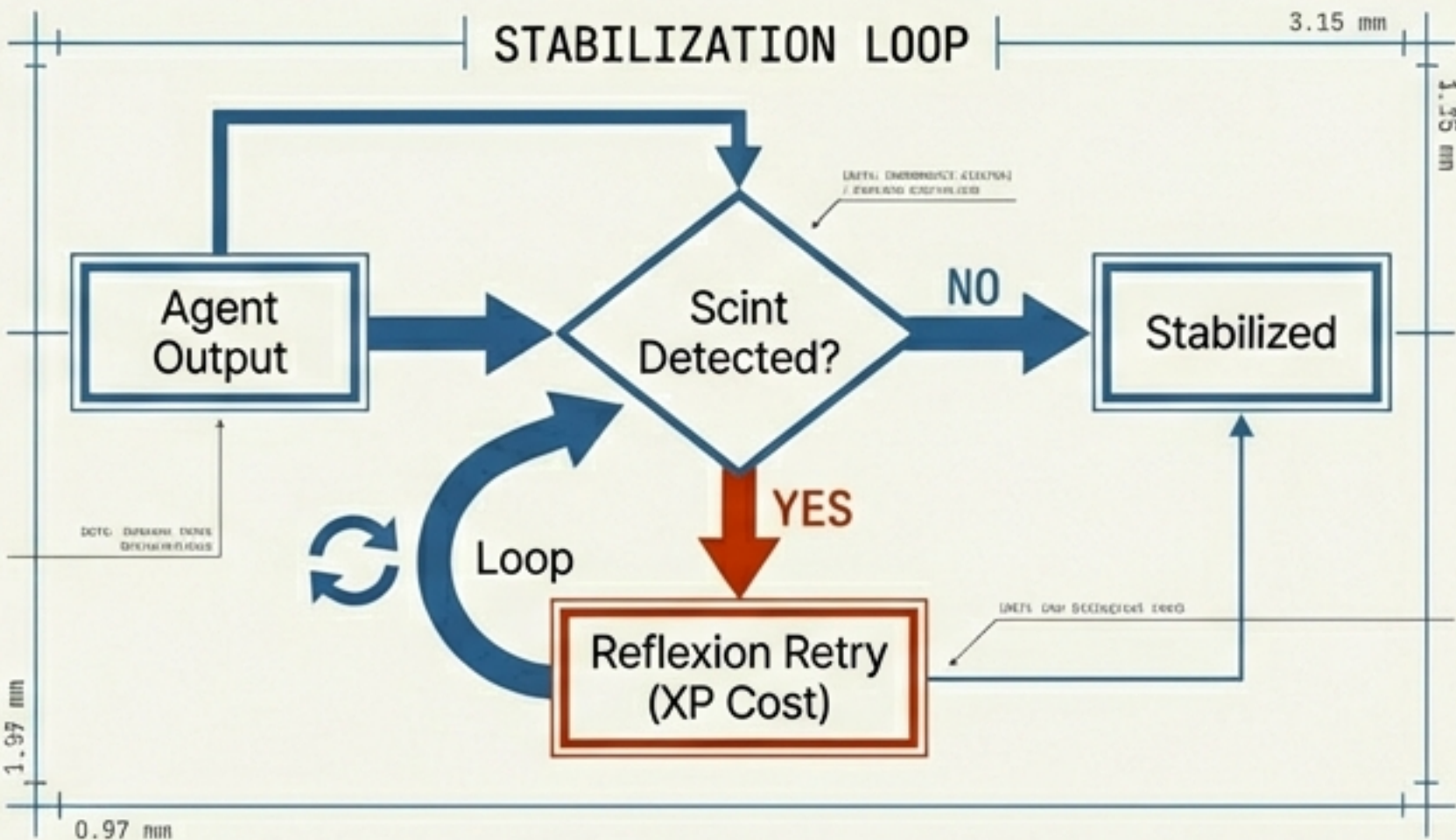
6.97 mm

CHARACTER SHEET / PYTHON CLASS DEFINITION

```
class ReliabilityAgent(RPGModel):
    def __init__(self):
        self.INT = 18 # Error Correction Rate
        self.WIS = 16 # Hallucination Check
        self.CHA = 14 # Prompt Adherence

    def roll_check(self, dc):
        # D20 + Mod vs DC
        return self.d20.roll() + self.get_mod() >= dc
```

INT (Intelligence):	Error Correction Rate	[18]
WIS (Wisdom):	Hallucination Check	[16]
CHA (Charisma):	Prompt Adherence	[14]



VERIFICATION DATA

Stability Index: 0.78 (High)

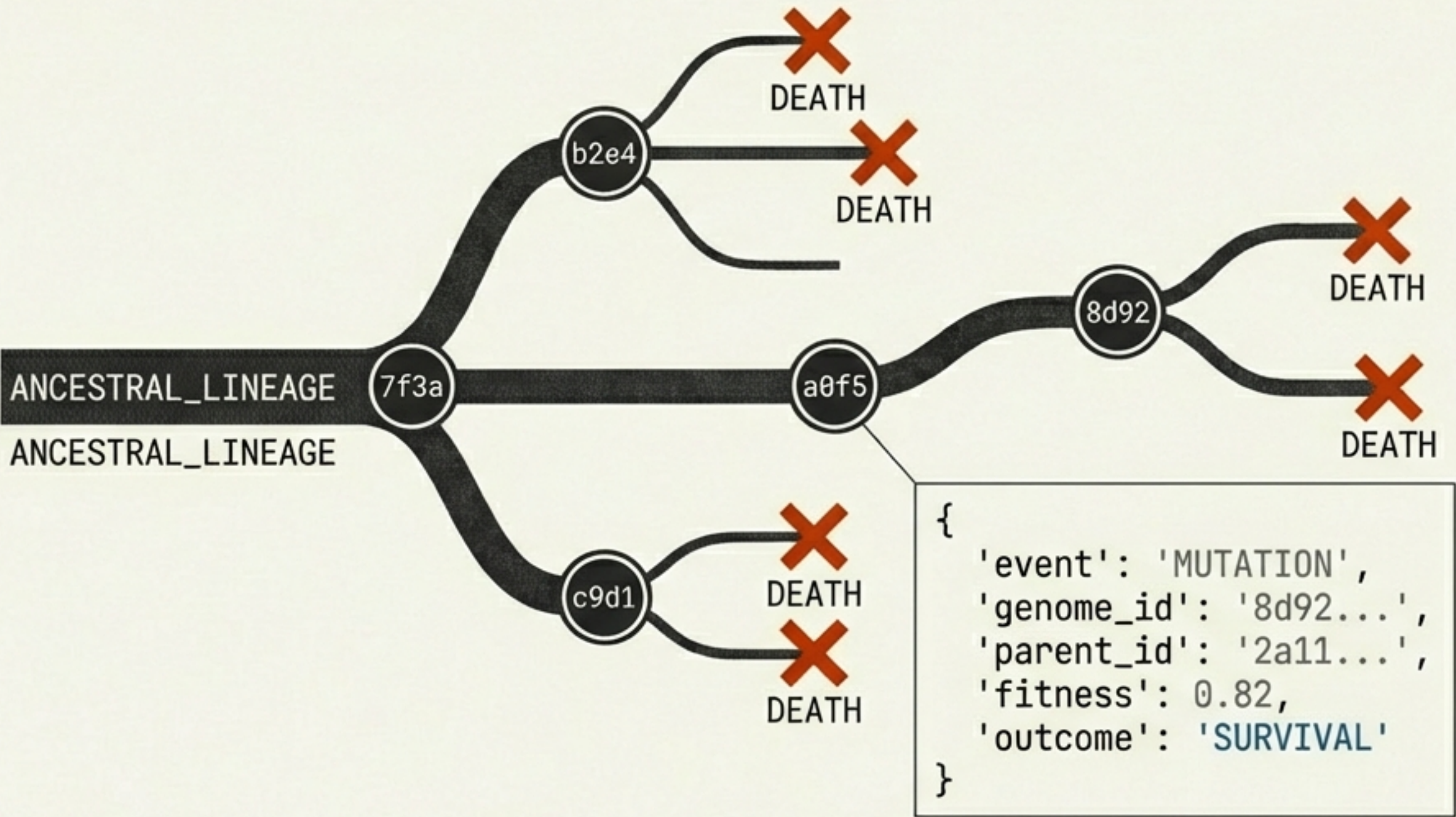
Critical Tests: 5/5 PASSING

Source: src/gym/rpg/

Verdict: "The most significant discovery of the audit."

Pillar 3: The Flight Recorder

Rigorous telemetry generating phylogenetic trees for scientific inquiry.



AUDIT STATUS:
85% OPERATIONAL

Data Volume:
35 JSONL files analyzed

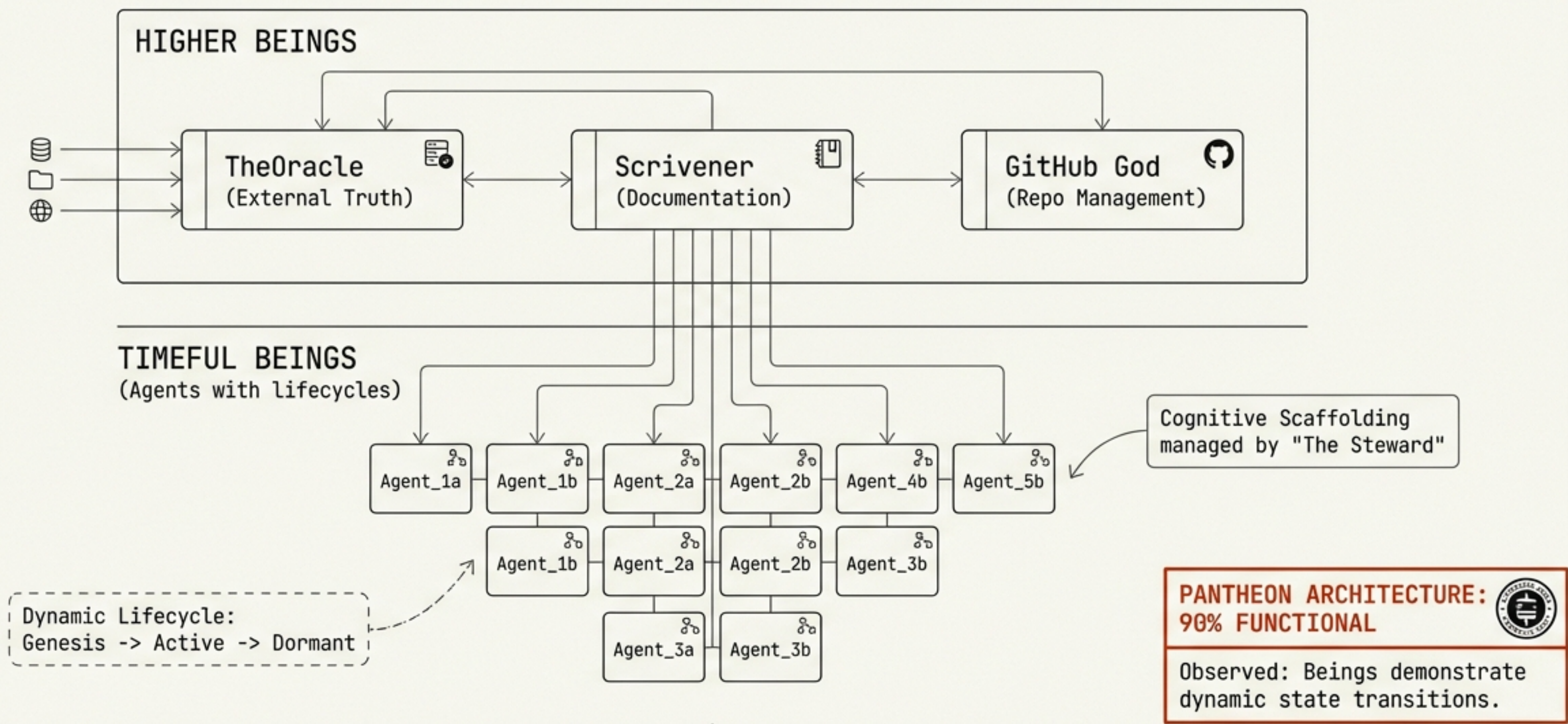
Total Rows:
964 lines of telemetry

Storage:
SQLite (waft_memory.db)

Storage:
SQLite (waft_memory.db)

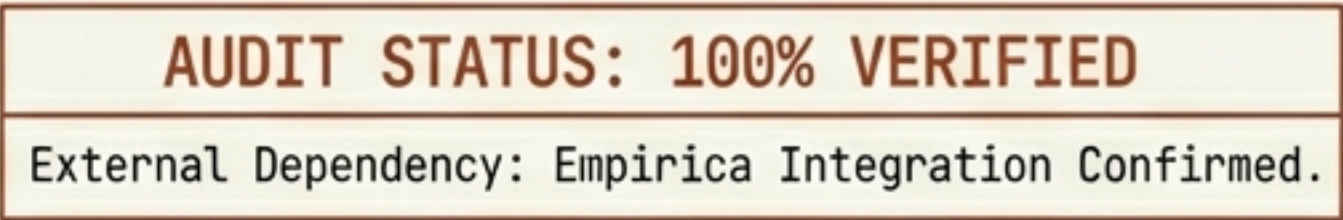
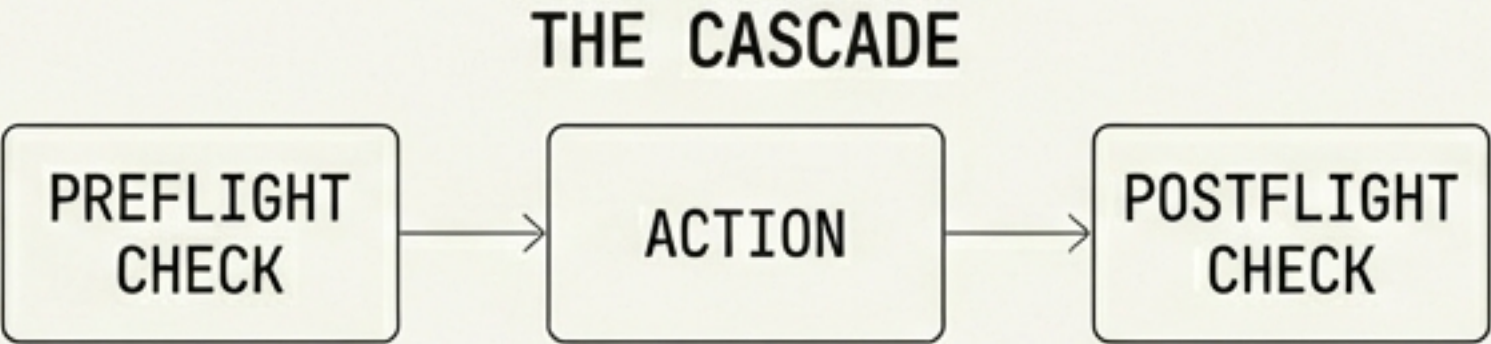
The Pantheon Architecture

A Society of Specialized Beings.



Epistemic Tracking: Knowing What We Know

Integration with Empirica for bounded autonomy and safety.



SAFETY GATES



PROCEED



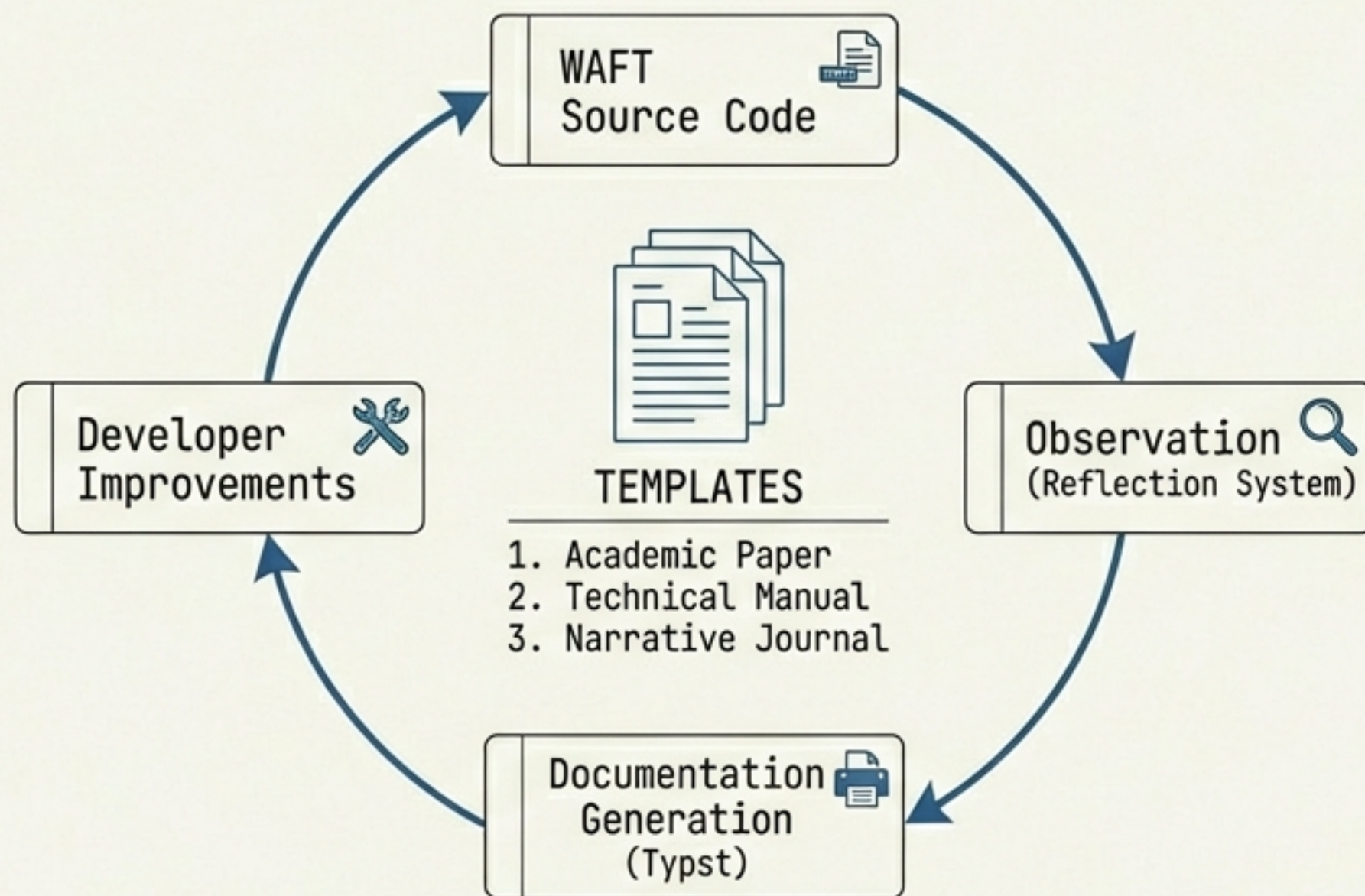
HALT



BRANCH

Recursive Self-Documentation

WAFT observes its own codebase to write its own manuals.



Verified in 'src/waft/reflection.py'. The system generates professional PDFs describing its own architecture.

Developer Experience: The CLI

How a researcher interacts with the laboratory.

```
$ waft new my_lab --path ./research
> Initializing evolutionary laboratory...
> Creating Pyrite memory structure...

$ waft spawn --agent RefactorBot --variants 10
> Spawning 10 variants...
> Genome IDs generated.

$ waft empirica monitor
> Launching TUI Dashboard...
```

Built with 'uv' package manager for modern Python dependency management.

Audit Findings: Reality vs. Placeholders

A transparent assessment of implementation gaps.

VALIDATED SYSTEMS (REAL)

- ✓ Genome Tracking (SHA-256)
- ✓ RPG Gym & Scint Mechanics
- ✓ Telemetry & Logging
- ✓ Pantheon Architecture
- ✓ Empirica Integration

IMPLEMENTATION GAPS (PLACEHOLDERS)

- ⚠ Evolutionary Cycle Automation (0% Complete)
- ⚠ Mutation Operators (40% - Partially Stubbed)
- ⚠ Multi-Agent Orchestration (50% - Limited)

TAKEAWAY: The engine is built, but the 'auto-pilot' logic is still in development.

Under the Hood: Technical Specifications

A detailed breakdown of the system's foundational technologies.

CORE LANGUAGE

Python 3.10+

Managed by "uv"



DATA PERSISTENCE

SQLite

waft_memory.db

_pyrite file structure



DOCUMENTATION ENGINE

Typst

Svelte



TESTING

pytest suite

380 Verified Tests



[DATA: WAFT_TECHNICAL_SPECS | STATUS: VALIDATED]

Project Structure: src/agents.py, src/gym/, src/waft/

The Final Verdict

LEGITIMATE & PROMISING

COMPLETENESS: 75% **STABILITY INDEX: 0.78** **STATUS: EXPERIMENTAL**

“A serious scientific instrument disguised as a developer tool. Suitable for research, but not yet for production.”

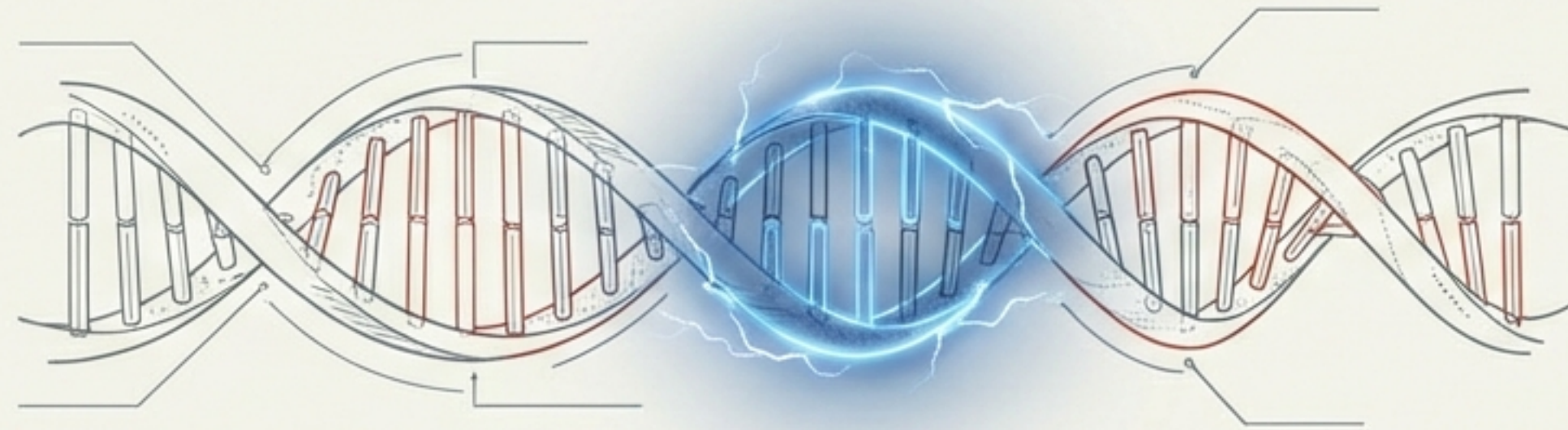
– Dr. Aria Vex, Systems Analyst

[DATA: FINAL_VERDICT_ASSESSMENT | STATUS: VALIDATED]

Project Structure: src/agents.py, src/gym/, src/

The Quest for the God-Head

The Roadmap to Artificial Cognition.



MISSION OBJECTIVES

1. **Immediate:** Close the Evolutionary Cycle loop (The "Coming Soon" feature).
2. **Goal:** Produce data for the book "The Physics of Artificial Cognition".

JOIN THE EXPERIMENT.

Clone the Repository: github.com/ctavolazzi/waft

[DATA: PROJECT_ROADMAP | STATUS: VALIDATED]

Evidence speaks louder than documentation.